Temperature Dependence of the Direct Band Gap of ZnSe and the Indirect Band Gap of Si

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We report the decrease in the band gap energies of ZnSe and Si with increasing temperature from 77 K to 300 K using near-infrared and visible light spectroscopy. The band gap energies were determined from Tauc plots. The temperature dependence of the band gaps was modelled using the Varshni, O'Donnell–Chen and Viña models. For both ZnSe and Si, the Varshni model yields systematically higher band gap values than the Viña model. However, the present experiment does not allow a definitive conclusion as to whether ZnSe, as a direct-gap semiconductor exhibits a larger temperature-induced band gap reduction than Si, an indirect-gap semiconductor over the measured temperature range.

I. INTRODUCTION

When atoms are brought into close proximity in a solid, the wavefunctions of their valence electrons spatially overlap and interact strongly [1]. The discrete atomic energy levels broaden to prevent electrons, forming valence and conduction bands separated by a forbidden energy gap. This phenomenon can be regarded as a crossover between atomic physics and condensed matter physics.

Semiconductors are materials in which the valence band is fully occupied as they possess an even number of valence electrons per primitive unit cell. the Fermi level lies at the valence band maximum [1, 2] and electronic excitation requires carriers to acquire sufficient energy to overcome the band gap. As a result, semiconductors exhibit lower electrical conductivity than metals, which lack a band gap and higher conductivity than insulators, which possess significantly larger band gaps [2].

Semiconductors are widely used in modern technological applications, owing to their moderate electrical conductivity and the tunability of their optical properties. For example, zinc selenide (ZnSe) has a wide band gap of 2.6 eV [3], making it a promising material for blue-green semiconductor diode lasers [4]. The band gap can be tuned by incorporating cadmium (Cd) and tellurium (Te) as dopants, forming the compound $Zn_{1-y}Cd_ySe_{1-x}Te_x$ [5]. By varying the composition of Cd, y and Te, x, the lasing wavelength of the laser can be precisely controlled.

Another notable property of semiconductors is the temperature dependence of the band gap. As temperature increases, the band gap energy decreases (discussed in detail in a later section), enabling the fabrication of temperature sensors for overheat protection in applications such as exhaust gas recirculation systems, CPUs, and GPUs [6]. This temperature-dependent tunability can also be exploited to adjust the laser wavelength, complementing the compositional method.

These tunable electronic and optical properties enables the design of materials with tailored electronic and optical properties for specific device applications. Therefore, understanding and characterising the optical properties of semiconductors is essential for the development and optimisation of modern semiconductor technologies. In this report, we study the band gap energies of ZnSe and silicon (Si), as well as the temperature dependence of their band gaps. A comparison between the two materials is also conducted to evaluate their respective suitability for different technological applications.

II. BACKGROUND

Optical absorption in semiconductors is governed by the structure of their electronic bands. In these materials, the valence band is fully occupied, and electrons must acquire sufficient energy to be promoted to the conduction band. As the forbidden band gap contains no available states, optical transitions require photons with energies exceeding the band gap.

Band gaps are classified as either direct or indirect. In a direct band gap semiconductor such as ZnSe, the valence-band maximum and conduction-band minimum occur at the same point in reciprocal k space, allowing electrons to undergo interband transitions solely by absorbing photons (shown in FIG. 1. In contrast, in indirect band gap semiconductors such as Si, these extrema occur at different k-values (shown in FIG. 2. When an electron is excited to the conduction band, a hole is left behind, and the Coulomb interaction between them forms an electron–hole pair.

As discussed in section I, semiconductors typically consist of atoms with an even number of valence electrons per unit cell, allowing the valence band to be completely filled. For this reason, group IV elements such as silicon (Si) and germanium (Ge), each possessing four valence electrons, form ideal elemental semiconductors. In addition to elemental semiconductors, compound semiconductors formed from elements symmetrically placed around group IV - such as gallium arsenide (GaAs) [1], which consists of group III and V elements - also exhibit semiconducting behaviour with well-defined band gaps. The size of the band gap is determined by the energy separation between bonding and antibonding molecular orbitals. In general, greater differences in electronegativity between constituent atoms lead to larger bonding–antibonding splitting, and thus larger band gap energies.

In the following sections, we will examine the theoretical optical properties of ZnSe and Si, including band gap energies and their temperature dependence, as well as optical density and absorption coefficients.

A. Direct Band Gap of ZnSe

ZnSe is a II–VI semiconductor compound, in which zinc atoms contribute two valence electrons and selenium atoms contribute four valence electrons, forming a fairly strong covalent crystal lattice. Since the electronegativity difference between zinc and selenium is greater than that between gallium and arsenic, ZnSe is expected to have a larger band gap energy than GaAs, which is 1.42 eV [8] due to the larger

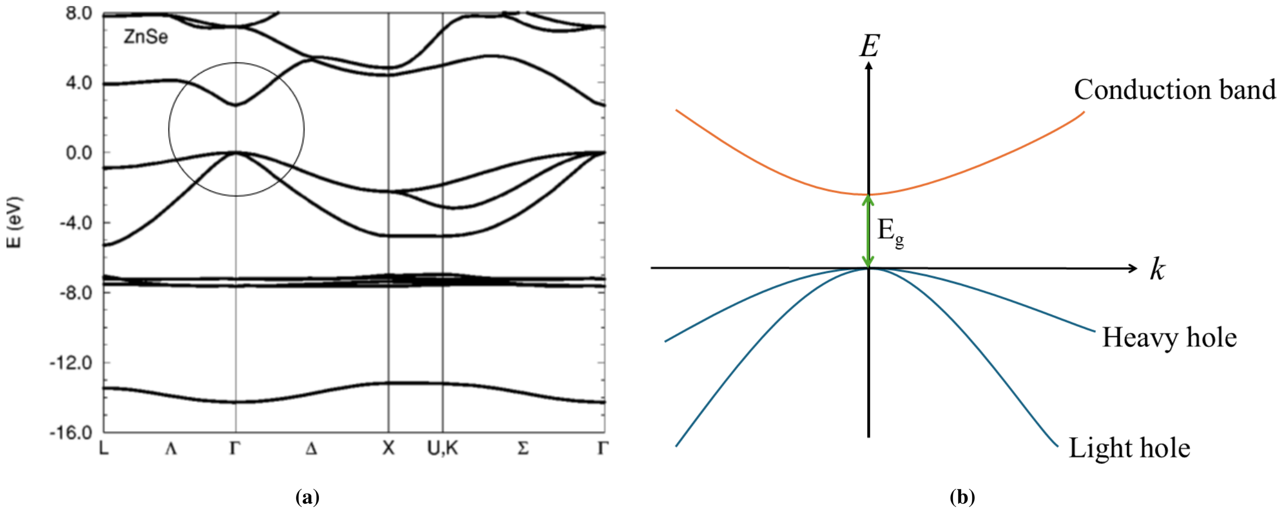


FIG. 1: Schematic diagram of the band structure of ZnSe. (a) Detailed band structure of ZnSe at room temperature obtained from [4]. The smallest band gap energy is determined at the Γ point in the circled region and has a value of 2.70 eV [7]. (b) Simplified band structure obtained by zooming in on the circled region in (a). The valence band maximum and conduction band minimum occur at the same crystal momentum k, indicating a direct band gap. Consequently, interband transitions can occur through photon absorption alone, without phonon assistance.

bonding–antibonding energy separation, ΔE.

FIG. 1 shows the band structure of ZnSe [4]. The band gap occurs at the Γ point in k-space. FIG. 1a shows the detailed band structure of ZnSe at different k values. The circled region is extracted and zoomed in to obtain the simplified band structure, as shown in FIG. 1b. It can be observed that the transition from the valence band (negative curvature) to the conduction band (positive curvature) is direct, with a band gap energy $E_g = 2.7$ eV [7]. The valence band consists of two sub-bands: the less negative curvature corresponds to the heavy hole, while the more negative curvature corresponds to the light hole. In principle, electrons with energy exceeding E_g can transition from either the heavy hole or light hole sub-band to the conduction band.

The band gap energy can be determined using various techniques such as photoelectron and inverse photoemission spectroscopy [9]. In this experiment, the Tauc plot method was employed, as it is more straightforward and is not subject to the limitations discussed in [9]. We first define absorption coefficient α, which can be obtained directly from the Beer–Lambert law:

$$\alpha = -\frac{1}{\ell} \ln \frac{I}{I_0} \tag{1}$$

where ℓ is the thickness of the sample, I is the transmitted light intensity and I_0 is the initial light intensity. In solids, the absorption coefficient can be very large - up to the order of 10^6 m^{-1} [1, 2]. This arises because solids have a high density of states.

Using the standard derivation of the density of states $g(k) = k^2/2\pi^2$ together with energy conservation during optical transitions, the density of states in the band is given by [1, 3]

$$g(\hbar\omega) = \begin{cases} 0, & \hbar\omega < E_g, \\ \frac{1}{2\pi^2}\left(\frac{2\mu}{\hbar^2}\right)^{3/2}(\hbar\omega - E_g)^{1/2}, & \hbar\omega \geq E_g. \end{cases} \tag{2}$$

where μ is the reduced electron-hole mass and $\hbar\omega$ is the photon energy.

From Fermi's golden rule, the optical absorption rate is proportional to the electronic density of states, as given in equation (2). For an allowed direct interband transition, the absorption coefficient for photon energies $\hbar\omega \geq E_g$ follows the Tauc relation (illustrated in Fig. 6),

$$\alpha_{\text{direct}}(\hbar\omega) = \mathrm{A}(\hbar\omega - E_g)^{1/2}, \tag{3}$$

where A is a proportionality constant. In the Tauc plot method, the quantity $(\alpha_{\text{direct}}\hbar\omega)^2$ is plotted as a function of $\hbar\omega$, which yields a linear relationship,

$$(\alpha_{\text{direct}}\hbar\omega)^2 = \mathrm{A}^2\hbar\omega - \mathrm{A}^2 E_g. \tag{4}$$

Photons with energies below E_g are transmitted through the material, while photons with energies exceeding E_g are absorbed. Consequently, when plotting the ratio of transmitted to incident light intensity, a sharp decrease is expected near the band gap, corresponding to the onset of absorption. As governed by equation (1), a decrease in this ratio results in an increase in the absorption coefficient α. As a result, the expected lineshape of a Tauc plot is, in theory, a sharp increase at the band gap, as indicated by the blue and orange lines in Fig. 6. According to equation (4), the linear portion of the Tauc plot can be extrapolated to the energy axis, and the band gap energy E_g is obtained from the x-intercept.

In the following sub-section, we will discuss the theoretical properties of silicon and how it, as an indirect-gap semiconductor, differs from ZnSe.

B. Indirect Band Gap of Si

Silicon is the most widely used semiconductor material in modern electronics due to its stability, abundance, and well-understood electronic properties. It has a band gap energy of 1.12 eV at room temperature [1, 11] and a shifted conduction band.

The photon wave vector is given by $k_p = 2\pi/\lambda$, where λ is the wavelength. For optical frequencies, $\lambda \sim 10^{-7}$[1],

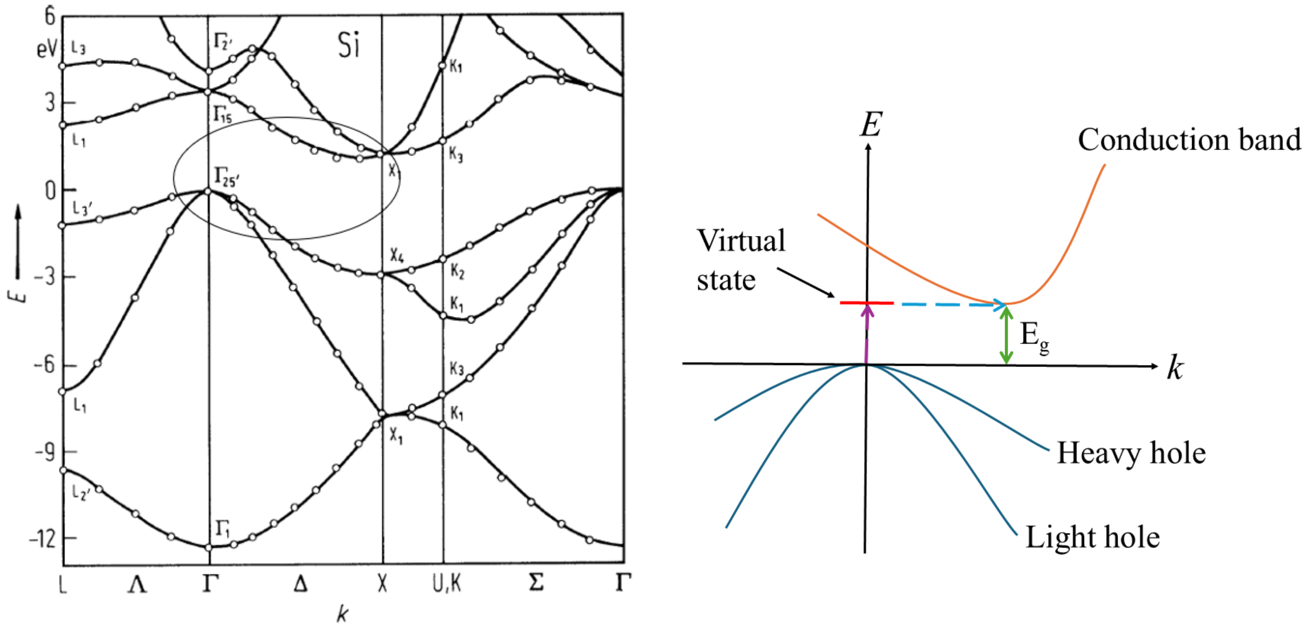


FIG. 2: Schematic diagram of band structure of Si. (a) The detailed band structure of Si obtained from [10]. The valence band is at Γ point whereas the conduction band has shifted to X point.The smallest indirect band gap energy is (b) Simplified band structure obtained by zooming in on the circled region in (a). The conduction band is shifted in k-space, requiring a photon (as indicated by purple dashed arrow) and a phonon in the transition (as indicated by blue dashed arrow). A virtual state is defined to derive equation (5).

giving a photon momentum of order 10^7. In contrast, electron wave vectors are approximately $k_e = \pi/a$ where $a \sim 10^{-10}$ is the lattice constant, yielding electron momenta of order 10^{10}[1]. Therefore, the momentum carried by photons is negligible compared to that of electrons. Since photons provide insufficient momentum to allow an indirect transition, a phonon must participate to conserve momentum, supplying the required shift in k space for the electron to reach the conduction band minimum.

As phonons are required in the transition, the absorption rate is lower than in direct-gap semiconductors such as ZnSe. This can be seen in the absorption-coefficient–energy graph shown in Fig. 3. For silicon, the absorption coefficient increases gradually with photon energy near the band gap, instead of rising sharply with a steep gradient as observed in direct-gap materials.

By defining a virtual state [3], where a photon excites an electron vertically to the conduction band (in Fig. 2b, the virtual state corresponds to the conduction band at $k = 0$) and applying energy conservation along with second-order perturbation theory [3], the Tauc relation for an allowed indirect transition with $\hbar\omega \geq E_g$ is

$$\alpha_{\text{indirect}}(\hbar\omega) = \mathrm{B}(\hbar\omega - E_g \pm \hbar\Omega)^2 \tag{5}$$

where B is the constant of proportionality, $\hbar\Omega$ is the energy of a phonon and the $\pm$ sign indicates phonon absorption or emission. In the Tauc plot method for an indirect-gap semiconductor, the quantity $(\alpha_{\text{indirect}}\hbar\omega)^{1/2}$ is plotted as a function of $\hbar\omega$, yielding the linear relationship

$$(\alpha_{\text{indirect}}\hbar\omega)^{1/2} = \sqrt{\mathrm{B}}(\hbar\omega - E_g \pm \hbar\Omega). \tag{6}$$

The band gap energy E_g is then determined using the same extrapolation procedure discussed in Section II A. Since the phonon energy $\hbar\Omega$ is typically much smaller than the

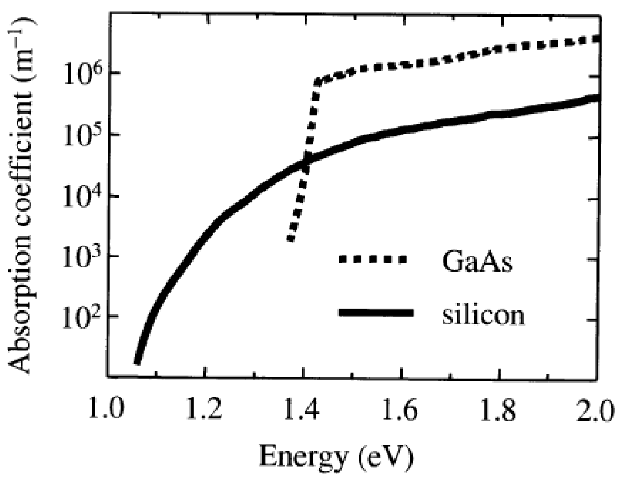


FIG. 3: A comparison of $\alpha - E$ graph between silicon and GaAs which is a direct-gap semiconductor. The electrons in silicon requires phonons in transition, which will slow the absorption rate. Thus, the absorption increases in a much steady rate near the band gap as compared to GaAs.

photon energy in the measured spectral range, it can be approximated as negligible, allowing E_g to be extracted from the x-intercept of the linear region. This result is consistent with the expectation that indirect-gap semiconductors exhibit lower absorption coefficients than direct-gap semiconductors, as indirect transitions have lower transition probabilities due to the requirement of phonon participation.

Phonon emission is possible at all temperatures, whereas phonon absorption occurs only when phonons are thermally populated [1]. In other words, we expect a temperature dependence in the observed band gap energy. This will be discussed in the following subsection.

C. Temperature Dependence of Band Gap Energy

As temperature increases, the crystal lattice of a semiconductor expands, increasing the average interatomic separation and reducing the strength of the electronic interactions that determine the band structure. This leads to a decrease in the band gap energy.

An empirical equation describing the effect of temperature on the band gap energy was proposed by Varshni [12] in 1967:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta} \tag{7}$$

where $E_g(0)$ is the band gap energy at 0 K, α and β are the material-dependent fitting parameters. The Varshni equation is applicable to both direct- and indirect-band-gap semiconductors. At high temperatures, the thermal expansion coefficient varies approximately linearly with T ; however, at low temperatures ($T < \Theta_D/10$, where Θ_D is Debye temperature), its dependence becomes nonlinear [12]. As a result, Varshni's equation deviates from experimental trends at low temperatures. Moreover, it lacks a rigorous theoretical foundation.

Another contributor to the temperature dependence of the band gap is the electron–phonon interaction, which is particularly significant for indirect band-gap semiconductors at elevated temperatures, as optical transitions require phonon participation. A full theoretical treatment via density functional theory approach are provided in [13]. In essence, thermal vibrations cause the nuclei to oscillate about their equilibrium positions, thereby introducing a perturbation to the Kohn-Sham potential. This perturbation renormalises the electronic band structure through two additional self-energy contributions: the Fan-Migdal and Debye-Waller terms, corresponding to the first- and second-order terms in the electron–phonon interaction, respectively.

Since this interaction depends on phonon population, it weakens at low temperatures as phonons lose their vibrational energies, a behaviour described by the Bose–Einstein distribution [1]:

$$f_{BE}(\hbar\Omega) = \frac{1}{e^{\hbar\Omega/k_B T} - 1}. \tag{8}$$

The resulting change in the band gap can be expressed as [3, 14]:

$$E_g(T) = E_g(0) - \mathrm{b}\left(1 + \frac{2}{e^{\theta/T} - 1}\right), \tag{9}$$

where b is the strength of interaction and θ is the energy of the phonons.

Donnell and Chen (1991) proposed a revision of Varshni's relation by incorporating equation (9). Their model takes the form [14]

$$E_g(T) = E_g(0) - \mathrm{S}\langle\hbar\omega\rangle\left[\coth\left(\frac{\langle\hbar\omega\rangle}{2k_B T}\right) - 1\right] \tag{10}$$

where S is a dimensionless coupling constant and $\langle\hbar\omega\rangle$ is the average phonon energy. Equation (9) can reduced to equation (10) by transforming the trigonometry to exponential form, where $\coth(x/2) = 1 + 2/(e^x - 1)$.

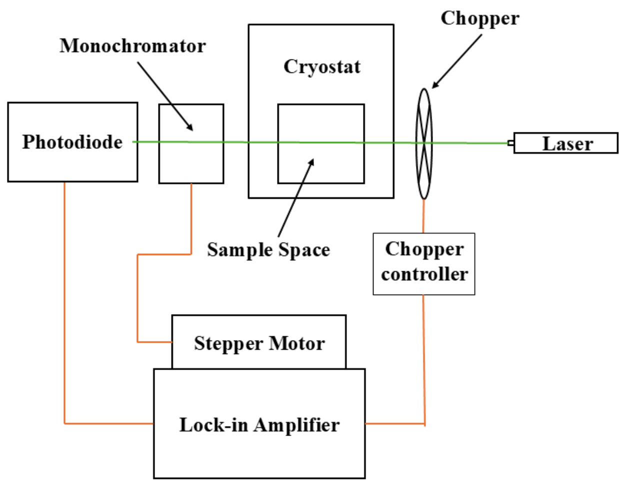


FIG. 4: Schematic diagram of the experimental setup used for the calibration of the monochromator. The laser beam was directed onto the photodiode via the chopper, monochromator and sample space. The chopper frequency was set to a prime number to prevent the lock-in amplifier from synchronising with ambient light. The optical path is indicated by the green line, while electrical connections are shown in orange.

III. METHODOLOGY

A. Callibration of Monochromator

In this experiment, an SMDC-04 monochromator was used, in which the wavelength is read from a four-digit mechanical analogue counter. However, the counter reading N is arbitrary; therefore, the instrument must be calibrated to establish a functional relationship between N and the light wavelength λ.

The calibration setup is shown in Fig. 4. Three helium-neon (He-Ne) lasers with colours red, yellow and green were used as laser sources possess significantly narrower linewidths than broadband white light. Each laser beam was directed through the monochromator and onto a photodiode, which converted the transmitted optical intensity into an electrical signal. The chopper frequency was set to a prime number to prevent the lock-in amplifier from synchronising with surrounding light fluctuations. The room was darkened and a series of electrical pulses was then sent to the stepper motor to drive the monochromator counter, enabling a scan across the wavelengths of the laser sources.

As the output of a gas laser exhibits inhomogeneous broadening (primarily thermal Doppler broadening due to the movements of the atoms experiencing different resonant frequencies), the recorded intensity–wavelength profiles are expected to follow Gaussian lineshapes. The mean of each Gaussian fit corresponds to the effective wavelength of that laser. When multiple scans were taken, the weighted mean of the fitted Gaussian centres was used to obtain a more accurate estimate of the effective wavelength.

B. Near Infrared and Visible Light Spectroscopy

The thicknesses of the samples were measured using a vernier calliper, and each sample was attached to the tip of the sample stick using minimal tape to avoid trapping air.

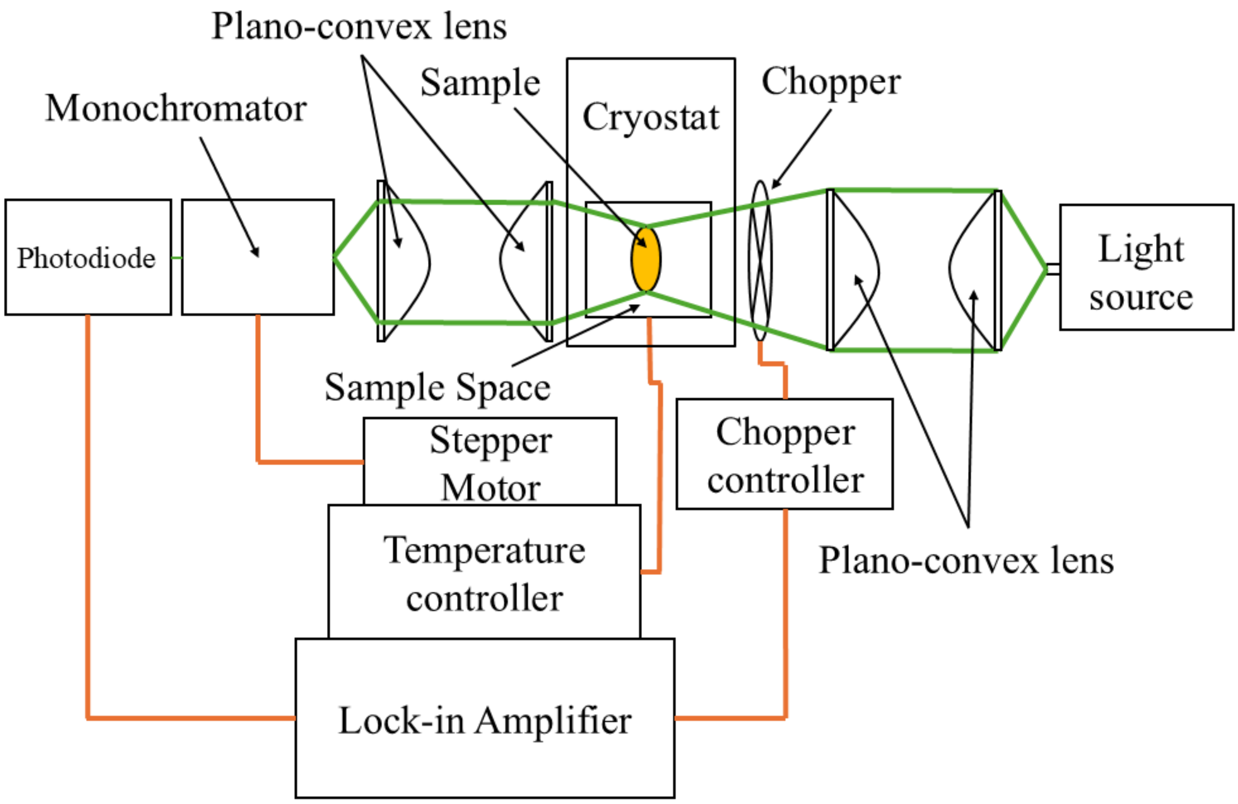


FIG. 5: Schematic diagram of the full setup for the measurements of I in varying temperatures. The optical path is indicated by the green lines, while electrical connections are shown in orange. The plano-convex lenses were used to collimate the lights.

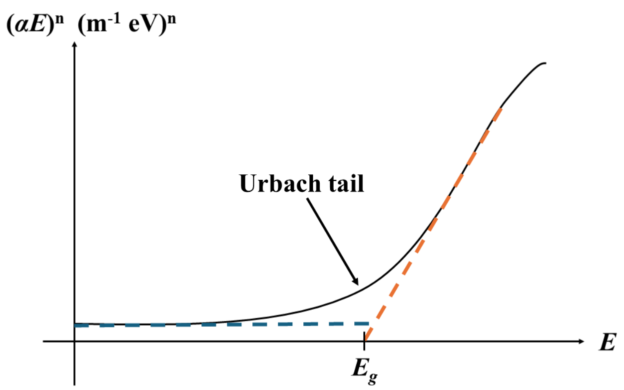


FIG. 6: Expected Tauc plot lineshape for allowed indirect ($n = 1/2$) and allowed direct ($n = 2$) interband transitions. In the ideal case, absorption begins sharply at the band gap energy, producing a horizontal region (blue dashed line) below the band gap and an immediate increase above it, as indicated by the orange dashed line. In practice, an exponential absorption region known as the Urbach tail [9] appears below the band gap.

Two pairs of plano–convex lenses were then tested to determine their focal lengths and positioned within the setup, with one pair placed between the monochromator and the sample space and the other between the chopper and the light source, in order to collimate the beams, as shown in Fig. 5. As the copper tip of the sample stick was slightly tilted, the light path deviated from a straight trajectory when travelling from the source to the monochromator. For this reason, we mounted the ZnSe sample onto the sample stick before the optical alignment was carried out.

Once the sample was in place, a laser source was used to align the lenses, sample space, monochromator, and photodiode. After the alignment was completed, the laser was replaced with a broadband white light source operating in the visible region. In this experiment, we used a Schott KL1500 illuminator, which provides an effective wavelength range within the visible spectrum.

The sample was then removed from the sample space, and the reference light intensity I_0 was measured by scanning the white light over the required wavelength range. Following this reference scan, the sample was reinserted, and the same wavelength range was scanned at temperatures from 77 K to 300 K.

A similar procedure was repeated for the silicon sample. The sample was mounted on the sample stick and the white light source was replaced with a Bentham IL1 illuminator to cover a broader wavelength range into the near-infrared. The reference light intensity was measured, after which the sample was reinserted and measurements were repeated at the same temperature range.

Tauc plots were constructed using equation (3) for a direct band-gap material and equation (5) for an indirect band-gap material. As indicated by the linear forms of equations (3) and (5), where the band gap E_g corresponds to the x-intercept, straight lines were fitted to the linear regions (orange dashed line as shown in FIG. 6 of the Tauc plots to determine the band gap energies of the samples.

IV. DISCUSSION

To construct Tauc plots, the time-domain data (x-axis) must first be converted into photon energy E. As the monochromator counter was set to decrease at a constant rate of 1.56 s per division (width of the pulses is 50 ms and a total of 1510 ms wait time when voltage is zero), the relationship between the counter reading N and time t is given by

$$N = N_0 - \frac{t}{5 \times 1.56} \tag{11}$$

where N_0 is the initial counter reading and t is the time taken to reach the maximum light intensity.

Intensity–time graphs were plotted for the red, yellow, and green lasers, as shown in FIG. 8a, b and c. The weighted means of the Gaussian fits provide three calibration points for the $N - \lambda$ relation, shown in FIG. 8d. A linear dependence between N and λ was observed and is described by

$$N = 1.006\lambda + 2270 \tag{12}$$

Substituting equation (12) into equation (11) yields the conversion from time to energy,

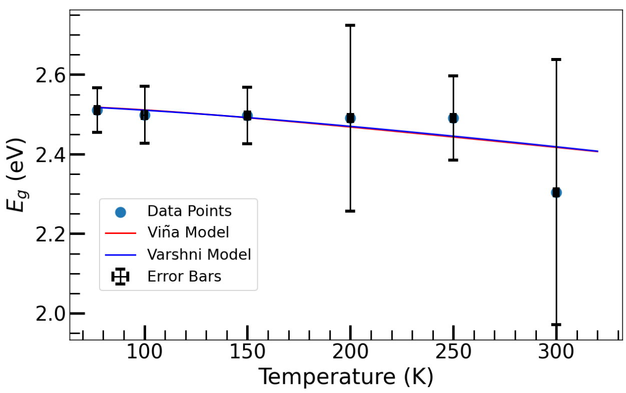
$$E = \frac{1.006hc}{N_0 - 0.128t - 2270} \tag{13}$$

where $N_0 = 2999$ for ZnSe, $N_0 = 3500$ for Si and $\lambda = hc/E$.

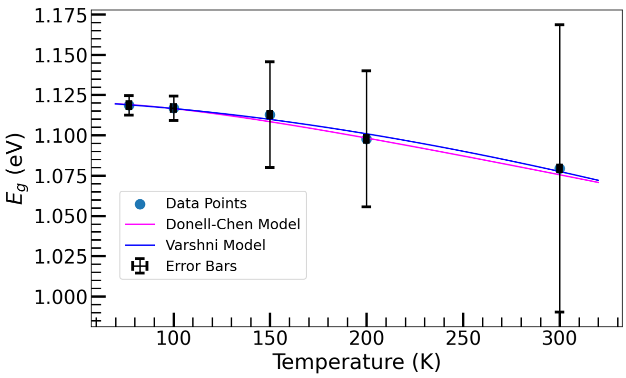
The band gap energies of ZnSe, $E_g^{(\mathrm{ZnSe})}$ and Si, $E_g^{(\mathrm{Si})}$, obtained from the Tauc plots are summarised in Table I. The temperature dependence of the band gap energies was modelled using equation (7) (Varshni model), equation (9) (phonon absorption and emission model), and equation (10) (O'Donnell-Chen model) and the resulting fits are shown in Fig. 7. Each of these models contains three fitting parameters. When all parameters were allowed to vary freely, the resulting uncertainties became very large, often exceeding the parameter values themselves, rendering the fits unreliable. This arises from the limited number of experimental data points, with only six measurements spanning the temperature range from 77 K to 300 K, which provides insufficient constraints on the models.

To address this limitation, all fitting parameters except $E_g(0)$ were fixed to literature values taken from [5, 12, 14]. A limitation of this approach is that samples of the same material may exhibit different parameter values due to variations in impurity concentration. Consequently, the true parameters for the samples used in this experiment may deviate from the adopted literature values.

The variation in $E_g^{(\mathrm{ZnSe})}$ (Fig. 7a) is small compared to that of $E_g^{(\mathrm{Si})}$ (Fig. 7b), as expected. The data point at 300 K exhibits a relatively large uncertainty due to an insufficient scanning range. Increasing the scanning range would shift the linear region of the Tauc plot to higher photon energies, resulting in a larger extracted value of $E_g^{(\mathrm{ZnSe})}$ at 300 K (as shown in FIG. 9). As a result, the temperature dependence of the ZnSe band gap can be approximated as nearly constant over the measured range. Consequently, the O'Donnell-Chen model is poorly constrained by the data and reliable values for the coupling constant S and the average phonon energy $\langle \hbar\omega \rangle$ of ZnSe cannot be extracted from the present measurements. Since no literature values for the



(a)



(b)

FIG. 7: The temperature dependence of the band gap energies of the materials. In principle, Viña and O'Donnell-Chen models fit better than Varshni given the same data [14] (a) Varshni and Viña models were used to fit the temperature dependence of the band gap and to extract the band gap energy of ZnSe at 0 K, yielding values of (2.529 ± 0.009) eV and (2.596 ± 0.009) eV, respectively. The variation in the band gap energy is negligibly small between 77 K and 250 K. The data point at 300 K exhibits a comparatively large uncertainty and a systematically lower value due to the shorter measurement duration, which reduced the number of data points in the linear region of the Tauc plot. (b) Varshni and O'Donnell-Chen models were used to extract the band gap energy of Si at 0 K, yielding values of (1.1222 ± 0.0002) eV and (1.1210 ± 0.0003) eV, respectively. Both values deviate from the accepted range of 1.16–1.17 eV [12, 14]. Both models agree with each other with a maximum deviation at 220 K of approximately 2 meV.

coupling constant of ZnSe are available, this model was replaced by Viña model.

The Varshni model shows good agreement with Viña model across the scanned temperature range, as shown in FIG. 7a. The Varshni parameters used were $\alpha = (7.3 \pm 0.4) \times 10^{-4}\,\mathrm{eV\,K^{-1}}$ and $\beta = (295 \pm 35)\,\mathrm{K}$, as reported in [5], yielding a best-fit value of $E_g(0) = (2.529 \pm 0.009)\,\mathrm{eV}$. For Viña model, the parameters $b = (73 \pm 4)\,\mathrm{meV}$ and $\theta = (260 \pm 10)\,\mathrm{K}$, taken from [5] were used, resulting in $E_g(0) = (2.596 \pm 0.009)\,\mathrm{eV}$. These results confirm that the band gap energy of ZnSe decreases with increasing temperature, consistent with the discussion presented in the Background section.

For silicon, both Varshni and O'Donnell–Chen models describe the temperature dependence of the band gap equally well within experimental uncertainty, with only minor deviations observed at intermediate temperatures. The

T (K)	$E_g^{(\mathrm{ZnSe})}$ (eV)	$E_g^{(\mathrm{Si})}$ (eV)	$-\Delta E_g^{(\mathrm{ZnSe})} (eV)$	$-\Delta E_g^{(\mathrm{Si})} (eV)$
77	2.513 ± 0.058	1.119 ± 0.007	–	–
100	2.501 ± 0.068	1.116 ± 0.007	0.012	0.003
150	2.500 ± 0.058	1.110 ± 0.007	0.001	0.006
200	2.499 ± 0.143	1.097 ± 0.046	0.001	0.013
250	2.493 ± 0.099	–	0.006	–
300	2.319 ± 0.298	1.078 ± 0.051	0.174	–

TABLE I: Band gap energies of ZnSe and Si at different temperatures, together with the corresponding changes in band gap energy ΔE_g between successive temperature points. A direct comparison of the temperature dependence of the band gap reduction in ZnSe and Si using the incremental values of ΔE_g is inconclusive, due to missing data points for Si at 250 K and the large uncertainty in $E_g^{(\mathrm{ZnSe})}$ at 300 K. Considering the overall change from 77 K to 300 K, ZnSe exhibits a larger band gap reduction (0.194 eV) than Si (0.041 eV); however, this comparison is strongly influenced by the high uncertainty in the ZnSe band gap at 300 K. Restricting the comparison to the temperature range 77 K to 200 K indicates a larger band gap reduction in Si than in ZnSe. Nevertheless, the limited number of data points in this temperature range is insufficient to draw a reliable quantitative conclusion.

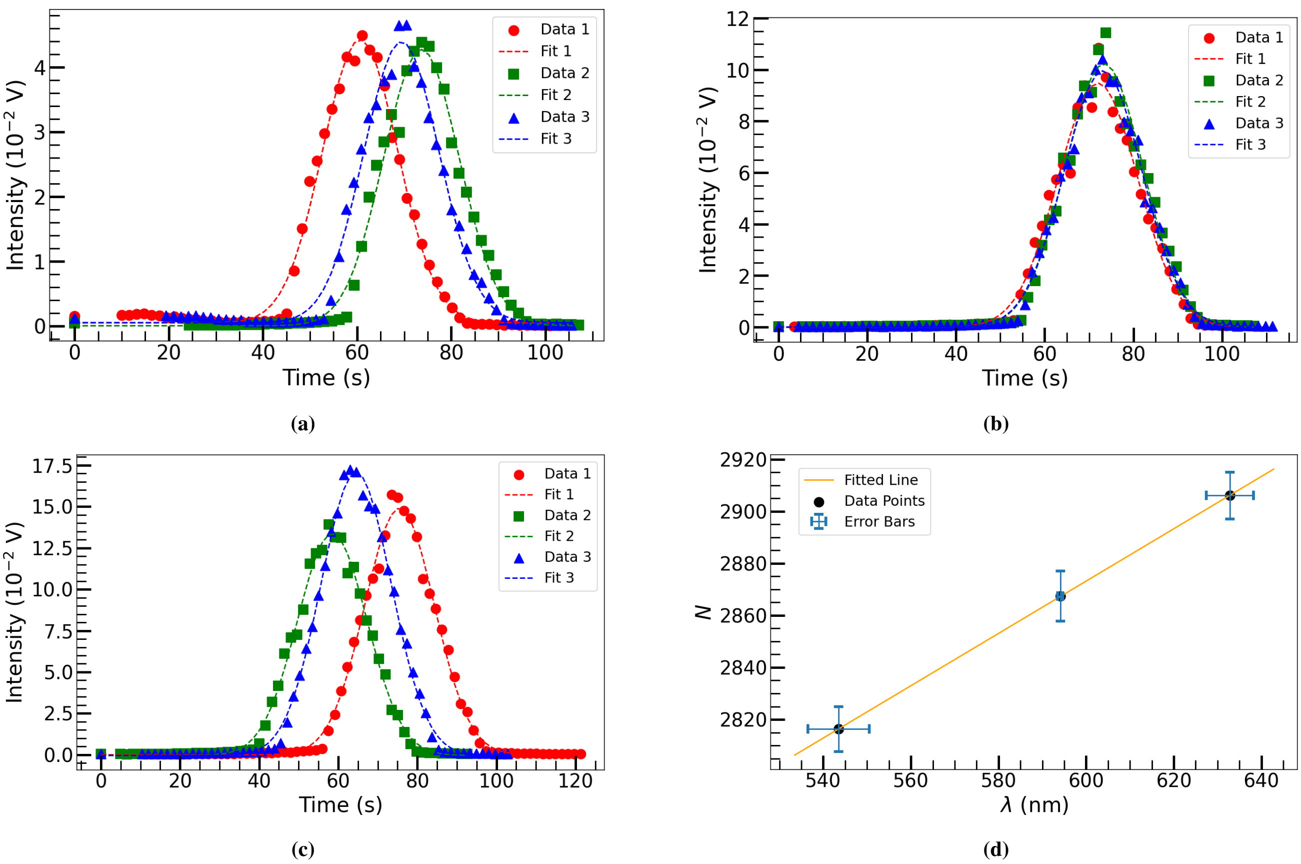


FIG. 8: Three measurements were taken for each laser, and the weighted mean was calculated by averaging the means of the corresponding Gaussian fits. The weighted means were plotted against time to determine the relationship between N and λ. (a) Red laser gives a weighted mean of (67 ± 5) s. (b) Yellow laser gives a weighted mean of (72.9 ± 0.6) s. (c) Green laser gives a weighted mean of (66 ± 7) s. (d) N is linearly related to λ with gradient $m = (1.005 \pm 0.003)$ nm^{-1} and intercept $c = 2269 \pm 2$.

Varshni parameters used were $\alpha = 7.021 \times 10^{-4}$ eV K^{-1} and $\beta = 1108$ K, as quoted in [12] without associated uncertainties, yielding a best-fit value of $E_g(0) = (1.1222 \pm 0.0002)$ eV. For O'Donnell–Chen model, the parameters $S = 1.49$ and $\langle\hbar\omega\rangle = 25.5$ meV, taken from [5] were used, resulting in a best-fit value of $E_g(0) = (1.1210 \pm 0.0003)$ eV. These values do not fully align with the commonly cited prediction that $E_g^{(\mathrm{Si})}$ lies in the range 1.16 to 1.17 eV [12, 14]. This discrepancy is likely due to the limited number of data points and the repeated measurements performed at each temperature. Nevertheless, the overall trend from 77 K to 300 K clearly demonstrates that the band gap energy of silicon decreases with increasing temperature, although a more comprehensive investigation would be required to accurately determine the value of $E_g^{(\mathrm{Si})}$ at 0 K.

Table I does not allow a definitive conclusion regarding which material exhibits a larger change in band gap energy over the temperature range 77 K to 300 K. Owing to the limited number of data points, two approaches may be used to quantify the temperature-induced band gap change: incremental changes between successive temperatures and the overall change across the full temperature range. The incre-

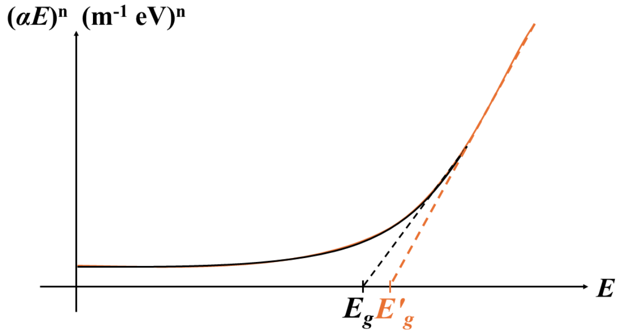


FIG. 9: In the experiment where the measurement duration was insufficient, the extracted band gap energy $E_g^{(300\mathrm{K})} = (2.319 \pm 0.298)$ eV is represented by the black curve and dashed line. This arises from the limited number of data points within the linear region of the Tauc plot. If the scanning range were extended to cover a broader wavelength range, the linear region would be more accurately defined, resulting in a blue-shift of the extracted band gap energy E_g', as indicated by the orange curve and dashed line.

mental approach does not yield a reliable comparison because the band gap energy of Si at 250 K is unavailable, making a consistent evaluation across all temperature intervals impossible.

When the overall change in band gap energy from 77 K to 300 K is considered, ZnSe appears to exhibit a larger reduction (0.194 eV) compared to Si (0.041 eV). However, this result is strongly influenced by the ZnSe data point at 300 K, which has a large uncertainty and yields a smaller band gap energy than expected, as discussed previously. To mitigate this effect, the comparison can be restricted to the temperature range 77 K to 200 K by excluding the data at 250 K and 300 K. Over this reduced range, Si exhibits a larger band gap reduction (0.022 eV) than ZnSe (0.014 eV). Nevertheless, given that this comparison relies on only four data points, the statistical significance is limited, and no robust quantitative conclusion can be drawn.

V. CONCLUSION

In summary, the band gap energies of both the direct-gap semiconductor ZnSe and the indirect-gap semiconductor Si decrease with increasing temperature. However, it is inconclusive whether the magnitude of the band gap reduction in Si is greater than that in ZnSe over the measured temperature range. Additionally, the absorption coefficient associated with the direct band gap of ZnSe is significantly larger than that of the indirect band gap of Si, consistent with the differing transition mechanisms in the two materials.

The observed reduction in the band gap energies of ZnSe and Si is consistent with lattice expansion and electron–phonon interactions, both of which renormalise the electronic band structure. However, the inability to distinguish conclusively between different temperature-dependent band gap models indicates that the available data are insufficient to constrain the fitting parameters reliably. The limited number of data points and the relatively large experimental uncertainties result in poor parameter determination. Parameters adopted from the literature may not accurately represent the samples used in this experiment, as these parameters are material- and sample-specific and can be influenced by factors such as impurity concentration and strain. Additional uncertainties arise from the the subjective identification of the linear region in the Tauc plots and mechanical backlash and slippage in the monochromator, all of which limit the precision of the extracted band gap energies.

Future measurements with finer temperature spacing and an extended temperature range would enable a more detailed investigation of the functional form of the temperature dependence and allow reliable determination of sample-specific model parameters.

To further quantify the factors influencing the determination of the band gap energies, the temperature dependence of the linewidths [5] of the direct gap of ZnSe and the indirect gap of Si could be analysed. This would allow investigation of intrinsic material effects, such as dislocation- and impurity-induced broadening as well as temperature-dependent mechanisms including electron–phonon interactions.

The high absorption coefficient of the direct band gap in ZnSe implies that radiative recombination is more probable and occurs without phonon assistance, resulting in a more efficient optical transition. Consequently, ZnSe is a more suitable candidate for optoelectronic applications such as light-emitting diodes and semiconductor lasers. In contrast, radiative recombination in indirect band gap Si is strongly suppressed, leading to longer carrier lifetimes. As a result, charge carriers remain mobile for extended periods within a conduction channel, making indirect band gap materials such as Si better suited for electronic applications, including the fabrication of metal–oxide–semiconductor field-effect transistors (MOSFETs).

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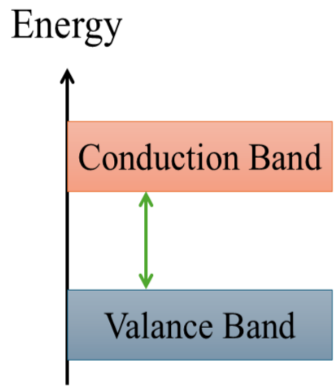
VII. LAY SUMMARY FOR GENERAL AUDIENCE

Materials are commonly classified as metals, semiconductors and insulators. From a quantum-mechanical perspective, their electrical behaviour can be understood using an energy-band picture. In this picture, electrons occupy a valence band, which contains the outermost electrons of atoms and a conduction band, where electrons are free to move and conduct electricity. These two bands are separated by an energy gap, known as the band gap (illustrated in Fig. 10). When electrons absorb enough energy such as from incoming light, they can overcome the band gap and move from the valence band into the conduction band.

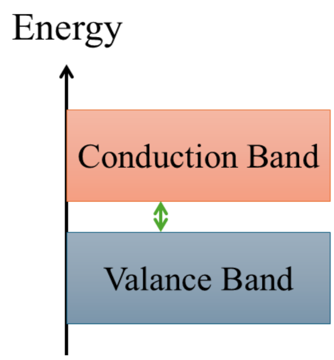
Semiconductors are particularly versatile materials for electronic and optical applications. For example, zinc selenide (ZnSe) can be used to fabricate blue-green lasers because its band gap corresponds to light in that colour range [5]. In addition, silicon (Si) is widely used in the fabrication of transistors, which are fundamental components in microchips found in mobile phones and computers.

This report investigates how the band gap energy of semiconductors depends on temperature. We first measured the band gap energies of zinc selenide and silicon at room temperature by shining white light onto the samples and identifying the energy at which light absorption begins. The samples were then cooled using liquid nitrogen to lower temperatures and the band gap energies were determined again using the same method.

We find that the band gap energy of a semiconductor decreases as temperature increases. This temperature sensitivity is an important and useful property, as it enables the development of electronic devices that respond to temperature changes. A common example is temperature monitoring in CPUs and GPUs, where semiconductor sensors help protect devices from overheating.



(a)



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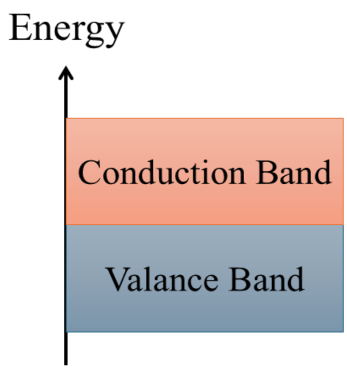

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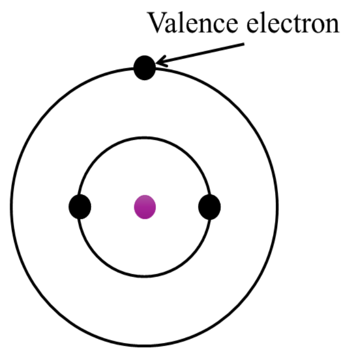

(d)

FIG. 10: Band structure of (a) insulators, which has the largest band gap energy (as indicated by the green double way arrow) (b) semiconductors, which has a smaller band gap than insulators (c) conductors, which do not have a band gap. (d) The outmost electron of an atom is called a valence electron circulating around a nucleus as indicated by purple dot.

VIII. GENERATIVE AI STATEMENT

In the production of this scientific paper, ChatGPT was used to correct grammar, improve the coherence of sentences, fix spelling mistakes, and enhance the flow of passages. The work and content are entirely my own, and the reference sources were selected without reliance on ChatGPT.